

# Liyan Chen

Web: <https://lly00412.github.io/>

Email : [lchen39@stevens.edu](mailto:lchen39@stevens.edu)

Mobile : +1-(201)9146333

## Education

- 
- |                                                                                                   |                                     |
|---------------------------------------------------------------------------------------------------|-------------------------------------|
| • <b>Stevens Institute of Technology (SIT)</b><br><i>Ph.D. in Computer Science</i>                | Hoboken, NJ<br>01/2020 - Present    |
| • <b>Stevens Institute of Technology (SIT)</b><br><i>M.S. in Information and Data Engineering</i> | Hoboken, NJ<br>08/2017 - 05/2019    |
| • <b>Renmin University of China (RUC)</b><br><i>B.S. in Mathematics</i>                           | Beijing, China<br>09/2012 - 07/2016 |

## Research Interests

- 
- |                                |                      |                    |
|--------------------------------|----------------------|--------------------|
| • Uncertainty Estimation       | • Semantic Mapping   | • Active Vision    |
| • Multi-View 3D Reconstruction | • Robotic Perception | • Machine Learning |

## Awards

- 
- |                                                                   |         |
|-------------------------------------------------------------------|---------|
| • AWS Machine Learning Research Awards (Prize with Sergül Aydıre) | 08/2019 |
|-------------------------------------------------------------------|---------|

## Publications

- 
- **Understanding while Exploring: Semantics-driven Active Mapping**  
Liyan Chen, Huangying Zhan, Hairong Yin, Yi Xu, Philippos Mordohai  
Neurips 2025, San Diego, USA.
  - **ActiveGAMER: Active GAussian Mapping through Efficient Rendering**  
Liyan Chen, Huangying Zhan, Kevin Chen, Xiangyu Xu, Qingan Yan, Changjiang Cai, Yi Xu  
CVPR 2025, Nashville, USA.
  - **Learning the Distribution of Errors in Stereo Matching for Joint Disparity and Uncertainty Estimation**  
Liyan Chen, Weihang Wang, Philippos Mordohai  
CVPR 2023, Vancouver, Canada.
  - **DropCluster: A Structured Dropout for Convolutional Networks**  
Liyan Chen, Philip Gautier, Sergül Aydıre  
(Oral) Women in Machine Learning workshop in NeurIPS 2019, Vancouver, Canada.

## Invited Talks

- 
- **Learning the Distribution of Errors in Stereo Matching for Joint Disparity and Uncertainty Estimation**  
NYC Computer Vision Day 2025, New York, USA
  - **DropCluster: A Structured Dropout for Convolutional Networks**  
Women in Machine Learning workshop in NeurIPS 2019, Vancouver, Canada

## Professional Experiences

- 
- |                                                                                                                                                                                                                                                                                  |                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|
| • <b>InnoPeak Technology</b><br><i>Internship</i> <ul style="list-style-type: none"><li>◦ Conducted collaborative research with other senior research staff and interns to design an advanced active mapping system for both geometric and photometric reconstruction.</li></ul> | Palo Alto, USA<br>05/2024 - 08/2024 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------|

- **ANBOUND Research Center** Beijing, China  
*Internship* 06/2016 - 10/2016
  - Carried out several consulting projects, including analysis of energy market data and automotive industry data, and advice on adjustments of petroleum product of our customer.
- **National Library of China** Beijing, China  
*Collaborating scholar* 07/2015 - 05/2024
  - Established the first comprehensive database for ancient documents from the National Library and five university libraries, and developed a framework to digitize, recognize rare characters, auto-classify topics, and simplify access for research and preservation.
- **National Laboratory of Data Engineering and Knowledge Engineering (RUC)** Beijing, China  
*Research Assistant* 08/2015 - 05/2016
  - Contributed to the Shandong Open Data Project, conducted research on the design of open data applications and sharing platforms.

---

## Professional Services

---

- |                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                   |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• <b>Teaching Assistant</b> <ul style="list-style-type: none"> <li>◦ EE551 Python Programing</li> <li>◦ CS383 Computer Organization and Programming</li> <li>◦ CS558 Computer Vision</li> <li>◦ CS559 Machine Learning</li> </ul> </li> </ul> | <ul style="list-style-type: none"> <li>• <b>Reviewer</b> <ul style="list-style-type: none"> <li>◦ CVPR 2024</li> <li>◦ ECCV 2024</li> <li>◦ CVPR 2025</li> <li>◦ ICRA 2026</li> </ul> </li> </ul> |
| <ul style="list-style-type: none"> <li>• <b>Seminar Organizer</b> <ul style="list-style-type: none"> <li>◦ Advanced Machine Learning Seminar at SIT</li> </ul> </li> </ul>                                                                                                           | <ul style="list-style-type: none"> <li>• <b>Workshop Facilitator</b> <ul style="list-style-type: none"> <li>◦ WiML Un-Workshop at ICML 2021</li> </ul> </li> </ul>                                |

---

## Open-Source Contributions

---

- **ActiveSGM: Semantics-driven Active Mapping**  
 Official implementation of ActiveSGM published at Neurips 2025.
- **ActiveGAMER: Active GAussian Mapping through Efficient Rendering**  
 Official implementation of ActiveGAMER published at CVPR 2024, internship project done with OPPO Research.
- **SEDNet: Stereo Error Matching Network**  
 Official implementation of SEDNet with GwcNet backbone published at CVPR 2023.
- **FLAML: A Fast Library for Automated Machine Learning & Tuning**  
 Contribute to designing the pipeline for automatic hyperparameter tuning in knowledge distillation using Transformer-based architectures.

---

## References

---

- |                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <ul style="list-style-type: none"> <li>• <b>Dr. Philippos Mordohai</b><br/>           Professor<br/>           Department of Computer Science<br/>           Stevens Institute of Technology<br/>           Email: pmordoha@stevens.edu</li> </ul> | <ul style="list-style-type: none"> <li>• <b>Dr. Enrique Dunn</b><br/>           Associate Professor<br/>           Department of Computer Science<br/>           Stevens Institute of Technology<br/>           Email: edunn@stevens.edu</li> </ul> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

- **Dr. Sergül Aydıre**  
Senior Applied Scientist  
Amazon AI  
Amazon Web Services  
Email: [sergulaydore@gmail.com](mailto:sergulaydore@gmail.com)

- **Dr. Huangying Zhan**  
Tech Lead Manager  
Goertek Alpha Labs  
Goertek Electronics  
Email: [zhanhuangying.work@gmail.com](mailto:zhanhuangying.work@gmail.com)